```
import pandas as pd
import io
from google.colab import files, output
from IPython.display import HTML, display

# 1. THE AUDIT LOGIC
def perform_audit_simple(contents):
    df = pd.read_excel(io.BytesIO(contents))

    # Audit stats
    total = len(df)
    missing = df.isnull().any(axis=1).sum()
    negatives = (df.select_dtypes(include=['number']) < 0).any(axis=1).sum()

    # Format rows for cards
    rows_html = ""
    for _, row in df.iterrows():
        d = row.to_dict()
        is_err = any(pd.isna(v) or (isinstance(v, (int, float)) and v < 0) for v in d.values())
        content = " | ".join([str(v) for v in d.values()])

        # Build individual cards directly in Python to avoid JS errors
        rows_html += f'''
        <div class="col-md-6">
            <div class="p-3 bg-white rounded shadow-sm border-start border-5 {'border-danger' if is_err else 'border-success'} ml
                <div class="d-flex justify-content-between align-items-center">
                    <small class="text-truncate" style="max-width: 70%;">{content}</small>
                    <span class="badge {'bg-danger' if is_err else 'bg-success'}">{'Review' if is_err else 'Verified'}</span>
                </div>
            </div>
        </div>'''

    return total, missing, negatives, rows_html

# 2. THE DASHBOARD DISPLAY
def show_dashboard(total=0, missing=0, neg=0, cards=""):
    html_ui = f"""
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
    <div class="container-fluid p-4" style="background: #f8fafc; min-height: 400px; border-radius: 15px;">
        <div class="p-5 text-center text-white mb-4 shadow" style="background: linear-gradient(135deg, #1e293b, #334155); border
            <h2 class="fw-bold">StealthAudit Intelligence</h2>
            <p class="mb-4">Internal Inventory & Data Auditing</p>
            <button onclick="document.getElementById('colab-upload-button').click()" class="btn btn-primary btn-lg px-5">Analyze
            <button onclick="window.print()" class="btn btn-outline-light btn-lg ms-2">Download PDF</button>
        </div>

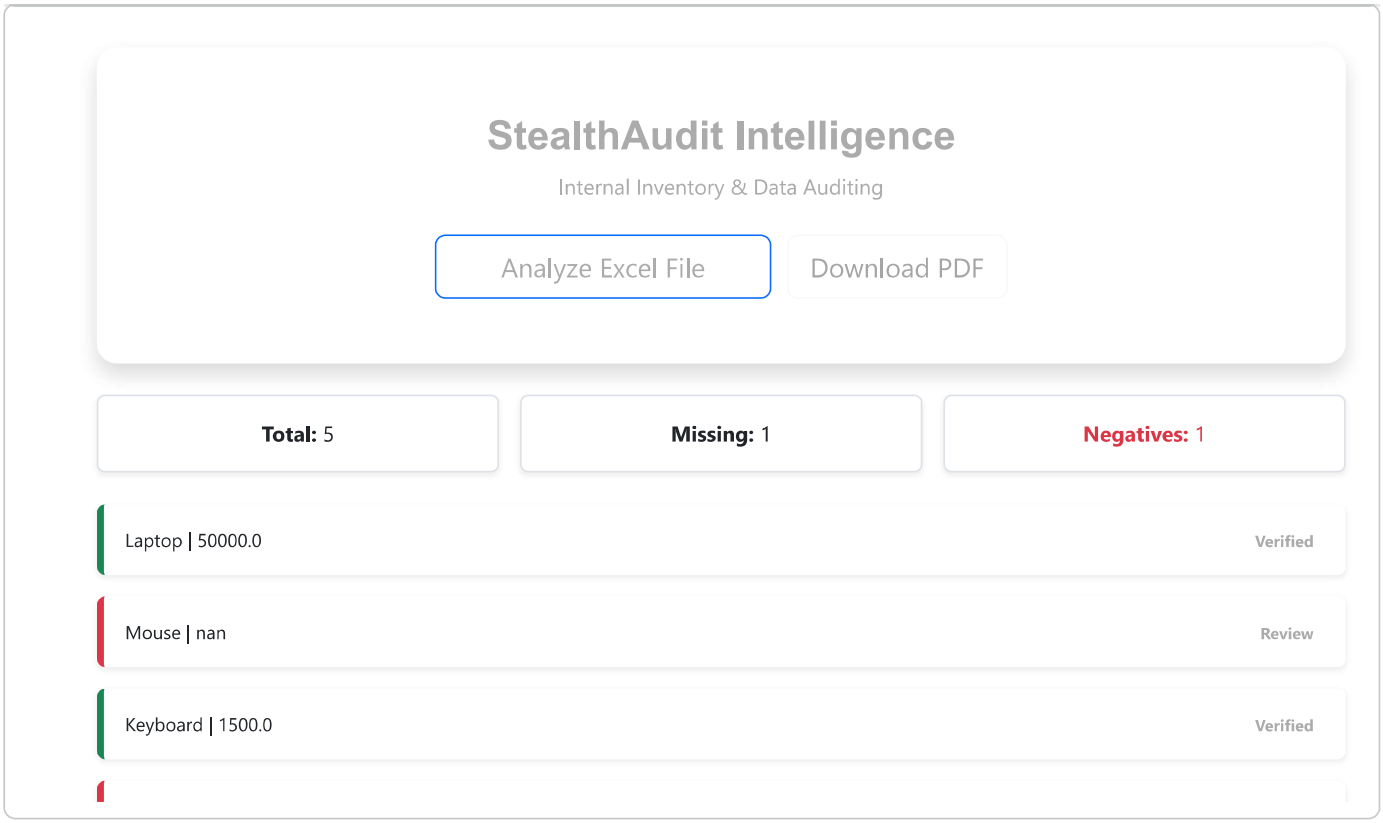
        <div class="row g-3 text-center mb-4">
            <div class="col-4"><div class="p-3 bg-white rounded shadow-sm border"><b>Total:</b> {total}</div></div>
            <div class="col-4"><div class="p-3 bg-white rounded shadow-sm border"><b>Missing:</b> {missing}</div></div>
            <div class="col-4"><div class="p-3 bg-white rounded shadow-sm border text-danger"><b>Negatives:</b> {neg}</div></div
        </div>

        <div class="row g-2">{cards}</div>
    </div>
    """
    display(HTML(html_ui))

# 3. RUNNING THE PROCESS
show_dashboard() # Show empty dashboard first

print("--- UPLOAD SECTION ---")
uploaded = files.upload() # Native Colab upload

for name, data in uploaded.items():
    output.clear() # Refresh screen with data
    t, m, n, c = perform_audit_simple(data)
    show_dashboard(t, m, n, c)
```




```
import pandas as pd
import io
from google.colab import files, output
from IPython.display import HTML, display

def perform_audit_simple(contents):
    df = pd.read_excel(io.BytesIO(contents))
    total = len(df)
    missing = df.isnull().any(axis=1).sum()
    negatives = (df.select_dtypes(include=['number']) < 0).any(axis=1).sum()

    rows_html = ""
    for _, row in df.iterrows():
        d = row.to_dict()
        is_err = any(pd.isna(v) or (isinstance(v, (int, float)) and v < 0) for v in d.values())
        content = " | ".join([str(v) for v in d.values()])

        # We add a 'data-search' attribute so the JS can find the text easily
        rows_html += f'''
        <div class="col-md-6 audit-item-card" data-search="{content.lower()}">
            <div class="p-3 bg-white rounded shadow-sm border-start border-5 {'border-danger' if is_err else 'border-success'} ml
                <div class="d-flex justify-content-between align-items-center">
                    <small class="text-truncate" style="max-width: 70%;">{content}</small>
                    <span class="badge {'bg-danger' if is_err else 'bg-success'}">{'Review' if is_err else 'Verified'}</span>
                </div>
            </div>
        </div>'''
    return total, missing, negatives, rows_html

def show_dashboard(total=0, missing=0, neg=0, cards=""):
    html_ui = f"""
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
    <div class="container-fluid p-4" style="background: #f8fafc; min-height: 400px; border-radius: 15px;">
        <div class="p-5 text-center text-white mb-4 shadow" style="background: linear-gradient(135deg, #1e293b, #334155); border
            <h2 class="fw-bold">StealthAudit Intelligence</h2>
            <p class="mb-4">Internal Inventory & Data Auditing</p>
            <div class="d-flex justify-content-center gap-2">
                <button onclick="document.getElementById('colab-upload-button').click()" class="btn btn-primary btn-lg">Analyze E
                <button onclick="window.print()" class="btn btn-outline-light btn-lg">Download PDF</button>
            </div>
        </div>

        <div class="row mb-4">
            <div class="col-12">
```

```
                <input type="text" id="searchInput" onkeyup="filterCards()" class="form-control form-control-lg shadow-sm" placel
            </div>
        </div>

        <div class="row g-3 text-center mb-4">
            <div class="col-4"><div class="p-3 bg-white rounded shadow-sm border"><b>Total:</b> {total}</div></div>
            <div class="col-4"><div class="p-3 bg-white rounded shadow-sm border"><b>Missing:</b> {missing}</div></div>
            <div class="col-4"><div class="p-3 bg-white rounded shadow-sm border text-danger"><b>Negatives:</b> {neg}</div></div
        </div>

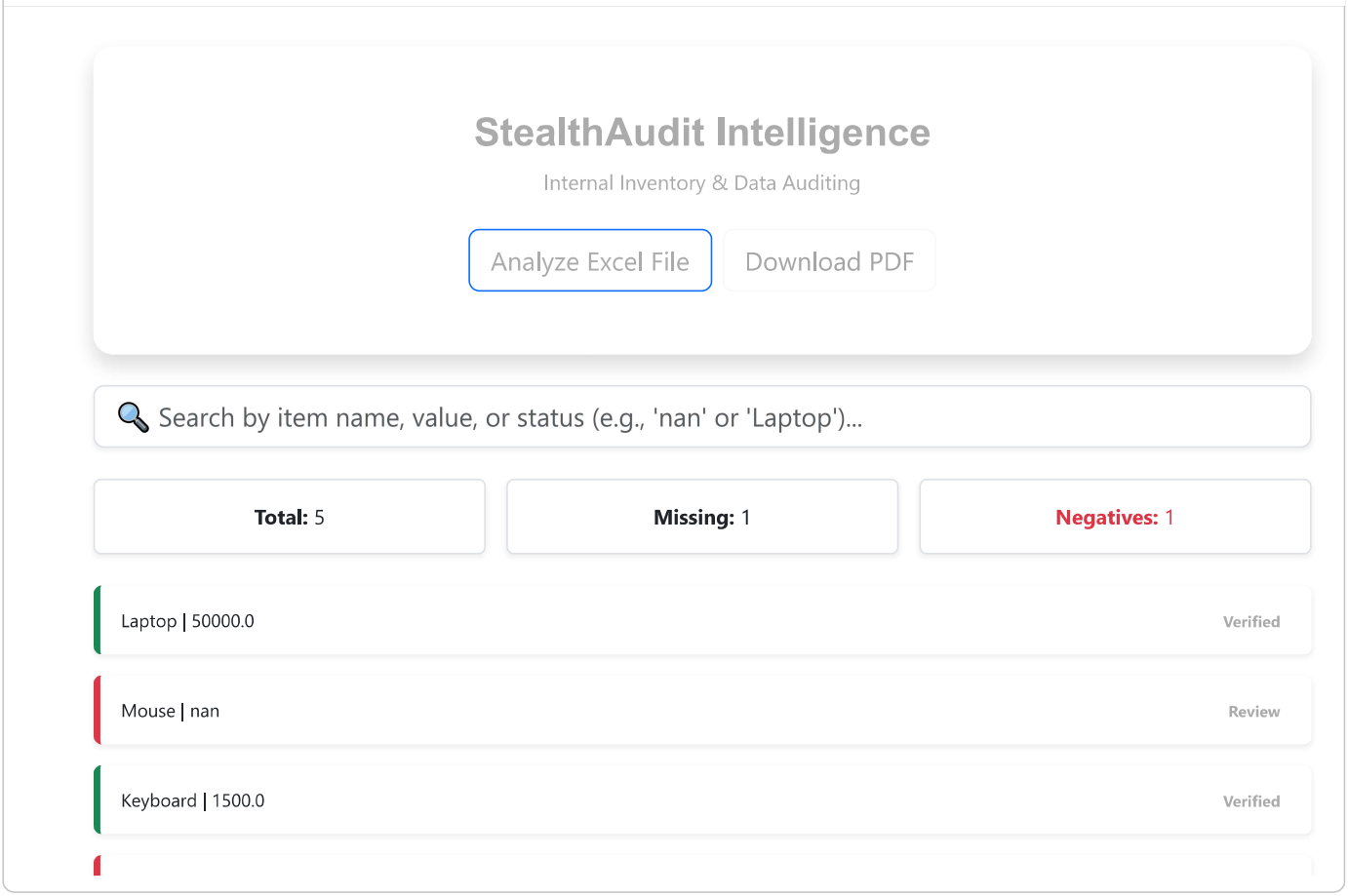
        <div class="row g-2" id="resultsGrid">{cards}</div>
    </div>

    <script>
        function filterCards() {{
            let input = document.getElementById('searchInput').value.toLowerCase();
            let cards = document.getElementsByClassName('audit-item-card');

            for (let i = 0; i < cards.length; i++) {{
                let content = cards[i].getAttribute('data-search');
                if (content.includes(input)) {{
                    cards[i].style.display = "";
                }} else {{
                    cards[i].style.display = "none";
                }}
            }}
        }}
    </script>
    """
    display(HTML(html_ui))

show_dashboard()
print("--- UPLOAD SECTION ---")
uploaded = files.upload()

for name, data in uploaded.items():
    output.clear()
    t, m, n, c = perform_audit_simple(data)
    show_dashboard(t, m, n, c)
```




```
import pandas as pd
```

```
import io
import base64
from google.colab import files, output
from IPython.display import HTML, display

def perform_audit_advanced(contents):
    df = pd.read_excel(io.BytesIO(contents))

    # 1. Audit Rules
    duplicates_count = df.duplicated(subset=[df.columns[0]]).sum()
    total = len(df)
    missing = df.isnull().any(axis=1).sum()
    negatives = (df.select_dtypes(include=['number']) < 0).any(axis=1).sum()

    # 2. Create Cleaned Data (Fixes negatives to 0 and fills missing with 'N/A')
    df_clean = df.copy()
    num_cols = df_clean.select_dtypes(include=['number']).columns
    df_clean[num_cols] = df_clean[num_cols].clip(lower=0)
    df_clean = df_clean.fillna("N/A")

    # Prepare Excel Download Link
    towrite = io.BytesIO()
    df_clean.to_excel(towrite, index=False, engine='openpyxl')
    towrite.seek(0)
    b64 = base64.b64encode(towrite.read()).decode()
    dl_link = f'data:application/vnd.openxmlformats-officedocument.spreadsheetml.sheet; base64,{b64}'

    # 3. Build Card UI
    rows_html = ""
    for i, row in df.iterrows():
        d = row.to_dict()
        has_data_error = any(pd.isna(v) or (isinstance(v, (int, float)) and v < 0) for v in d.values())
        is_duplicate = df[df.columns[0]].duplicated(keep=False).iloc[i]
        content = " | ".join([str(v) for v in d.values()])

        if has_data_error:
            status, border = '<span class="badge bg-danger">⚠️ Review</span>', "border-danger"
        elif is_duplicate:
            status, border = '<span class="badge bg-warning text-dark">🚩 Duplicate</span>', "border-warning"
        else:
            status, border = '<span class="badge bg-success">✅ Verified</span>', "border-success"

        rows_html += f'''
        <div class="col-md-6 audit-item-card" data-search="{content.lower()}">
            <div class="p-3 bg-white rounded shadow-sm border-start border-5 {border} mb-2">
                <div class="d-flex justify-content-between align-items-center">
                    <small class="text-truncate" style="max-width: 70%; font-size: 0.85rem;">{content}</small>
                    {status}
                </div>
            </div>
        </div>'''
    return total, missing, negatives, int(duplicates_count), rows_html, dl_link

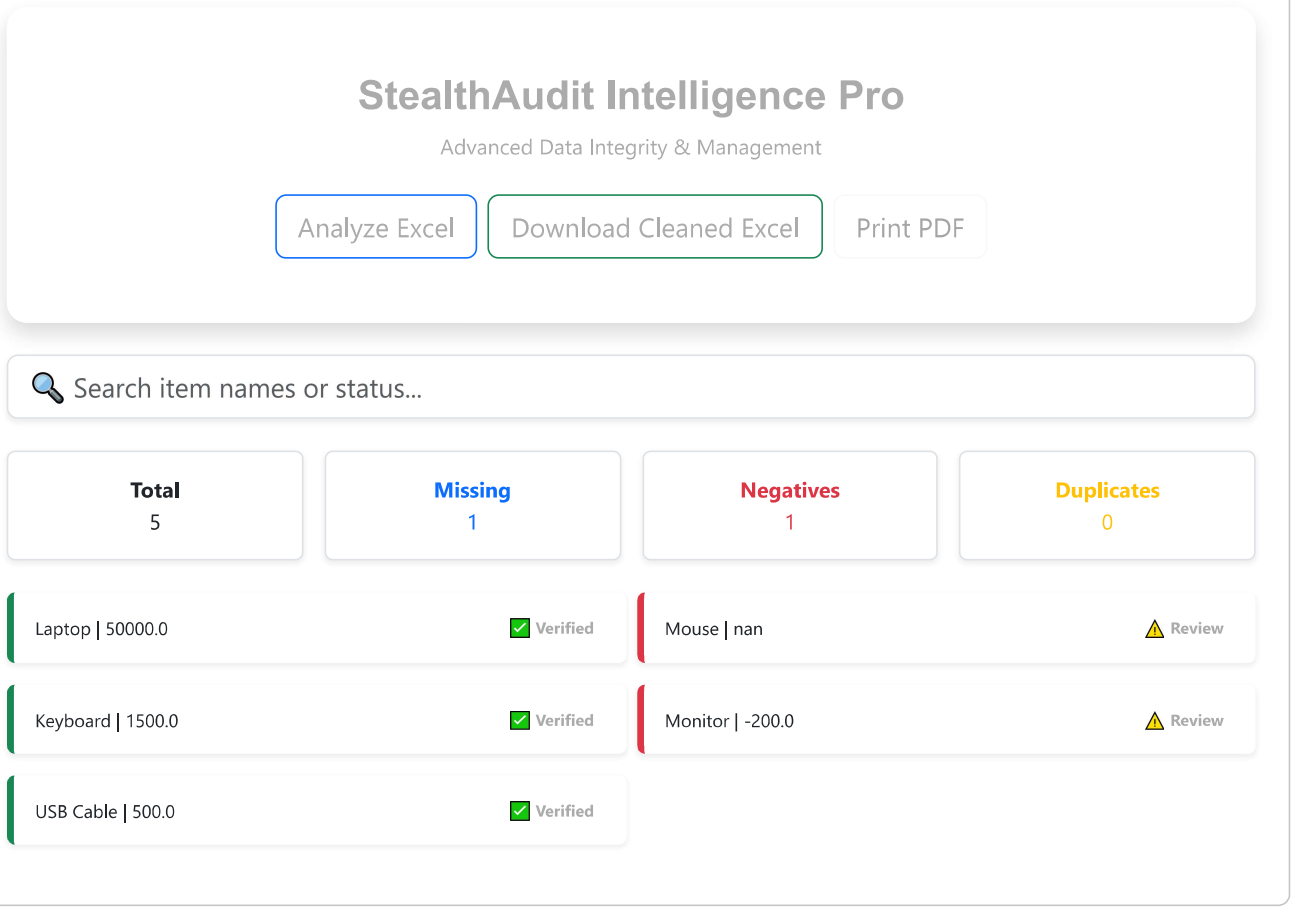
def show_dashboard(total=0, missing=0, neg=0, dups=0, cards="", dl="#"):
    html_ui = f"""
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
    <div class="container-fluid p-4" style="background: #f8fafc; min-height: 500px; border-radius: 15px;">
        <div class="p-5 text-center text-white mb-4 shadow" style="background: linear-gradient(135deg, #1e293b, #334155); border
            <h2 class="fw-bold">StealthAudit Intelligence Pro</h2>
            <p class="mb-4">Advanced Data Integrity & Management</p>
            <div class="d-flex justify-content-center gap-2">
                <button onclick="document.getElementById('colab-upload-button').click()" class="btn btn-primary btn-lg">Analyze
                <a href="{dl}" download="Cleaned_Audit_Report.xlsx" class="btn btn-success btn-lg {'disabled' if dl=='#' else ''}
                <button onclick="window.print()" class="btn btn-outline-light btn-lg">Print PDF</button>
            </div>
        </div>

        <input type="text" id="searchInput" onkeyup="filter()" class="form-control form-control-lg shadow-sm mb-4" placeholder="

        <div class="row g-3 text-center mb-4">
            <div class="col-3"><div class="p-3 bg-white rounded shadow-sm border"><b>Total</b><br>{total}</div></div>
            <div class="col-3"><div class="p-3 bg-white rounded shadow-sm border text-primary"><b>Missing</b><br>{missing}</div>
            <div class="col-3"><div class="p-3 bg-white rounded shadow-sm border text-danger"><b>Negatives</b><br>{neg}</div></d
            <div class="col-3"><div class="p-3 bg-white rounded shadow-sm border text-warning"><b>Duplicates</b><br>{dups}</div>
        </div>
        <div class="row g-2" id="resultsGrid">{cards}</div>
    </div>
    <script>
```

```
        <script>
            function filter() {{
                let val = document.getElementById('searchInput').value.toLowerCase();
                let items = document.getElementsByClassName('audit-item-card');
                for (let i = 0; i < items.length; i++) {{
                    items[i].style.display = items[i].getAttribute('data-search').includes(val) ? "" : "none";
                }}
            }}
        </script>
        """
        display(HTML(html_ui))

show_dashboard()
print("--- UPLOAD SECTION ---")
uploaded = files.upload()
for name, data in uploaded.items():
    output.clear()
    t, m, n, d, c, dl = perform_audit_advanced(data)
    show_dashboard(t, m, n, d, c, dl)
```




```
import shutil
# Replace 'my_project_folder' with the name of your folder in Colab
# If your files are in the main directory, use '/content'
shutil.make_archive('StealthAudit_Project', 'zip', '/content')
```

```
'/content/StealthAudit_Project.zip'
```

```
from google.colab import files
files.download('StealthAudit_Project.zip')
```